

AI-Based Infant Cry Diagnostic System: Phase 3 — Hybrid Feature Fusion Technical Validation & Results Analysis

InfantCryProject

Advance Machine Learning & Deep Learning

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Abstract—This paper presents Phase 3 of an AI-based infant cry diagnostic system targeting Neonatal Intensive Care Unit (NICU) deployment. Phase 3 introduces a Hybrid Feature Fusion architecture combining deep spectral representations from a fine-tuned Convolutional Neural Network (CNN) with a parallel 70-dimensional hand-crafted acoustic feature branch. All experiments were conducted on the *Baby Crying Sounds* dataset published on Kaggle by Ahmed [1], comprising 1,197 labelled WAV files across 9 physiological and acoustic classes. Strategic class merging (`cold_hot` into `hungry`) and SMOTE oversampling were applied prior to training. Phase 3 achieves **42%** overall validation accuracy (Phase 2 baseline: 40.42%), with the Hungry class F1-score recovering from 0.08 (Phase 3 v1) to 0.39 via pitch trajectory feature injection. Laugh and Silence maintain perfect F1 scores of 1.00. Macro-average F1 reaches 0.47. Persistent confusion within the Hungry/Tired/Discomfort cluster motivates a Phase 4 transition to LSTM-based temporal sequence modelling.

Index Terms—infant cry classification, hybrid feature fusion, CNN, MFCC, Log-Mel spectrogram, SMOTE, NICU, audio classification, deep learning.

I. Introduction

Neonatal Intensive Care Unit (NICU) environments face a persistent caregiver gap: physiological distress signals from preterm infants — hunger, pain, discomfort, temperature dysregulation — manifest as cry vocalisations that are difficult for non-expert caregivers to distinguish reliably [5]. An automated, real-time cry classification system could augment caregiver response and reduce time-to-intervention for critical physiological states.

Prior work has demonstrated two broad approaches to infant cry classification: classical machine learning on hand-crafted acoustic features [3, 4], and end-to-end deep learning on spectral image representations [2]. Each approach suffers from a distinct limitation — classical methods lack complex spectral pattern recognition, while CNN methods lose explicit temporal texture cues such as pitch trajectory and energy envelope dynamics.

This paper describes Phase 3 of a multi-phase research project addressing this gap through a *Hybrid Feature Fusion* architecture: a Log-Mel Spectrogram is processed by a CNN backbone in parallel with a 70-dimensional hand-crafted acoustic feature vector, and both branches are fused before the classifier head.

Phase progression: Phase 1 established a classical ML baseline. Phase 2 deployed a 3-block CNN on Log-Mel Spectrograms (40.42% accuracy), identifying spectral confusion between tonally similar cry states as the primary bottleneck. Phase 3 (this report) addresses that bottleneck through hybrid fusion, pitch feature injection, and strategic class merging.

II. Dataset

A. Baby Crying Sounds Dataset

All experiments were conducted on the *Baby Crying Sounds* dataset published on Kaggle by Ahmed [1]. The dataset consists of **1,197 WAV audio files** sampled at 16 kHz, each clip approximately 7 seconds in duration, labelled across 9 physiological and acoustic states: `belly_pain`, `burping`, `cold_hot`, `discomfort`, `hungry`, `laugh`, `silence`, `tired`, and `noise`.

Table 1 summarises the raw class distribution, the strategic merge applied, and the post-SMOTE balanced counts.

Table 1. Baby Crying Sounds Dataset Composition [1]

Class	Raw	%	Post-SMOTE
Laugh	382	31.9	382
Hungry	135	11.3	382
Tired	132	11.0	382
Belly Pain	124	10.4	382
Burping	108	9.0	382
Silence	108	9.0	382
Discomfort	100	8.4	382
Cold/Hot	96	8.0	→ <i>Hungry</i>
Noise	0	0.0	<i>Excluded</i>
Total	1,197		3,056

B. Strategic Class Merging

Phase 3 v1 confusion matrix analysis revealed that `cold_hot` and `hungry` were misclassified at extremely high rates (33 of 76 test hungry samples predicted as `cold_hot`). Acoustic analysis of both classes in the Baby Crying Sounds dataset confirms that they share: (i) sustained, rhythmic cry patterns with similar periodicity; (ii) overlapping fundamental frequencies (250–500 Hz); and

(iii) near-identical Mel spectrogram contours. The merge is physiologically justified as both states represent unmet needs expressed through sustained vocalisation [5].

C. SMOTE Oversampling

Class imbalance (Laugh: 382 vs. Discomfort: 100 samples, a $3.8\times$ ratio) was addressed by applying SMOTE [3] to the 70-dimensional hand-crafted feature space:

$$x_{\text{syn}} = x_i + \lambda \cdot (\hat{x}_j - x_i), \quad \lambda \in [0, 1] \quad (1)$$

where x_i is a minority-class sample and $\hat{x}_j$ is one of its $k=5$ nearest neighbours. All classes were equalised to 382 samples (maximum increase $+282\%$ for Discomfort).

III. Phase 3 Architecture

A. Hybrid Feature Fusion Overview

The core motivation is that spectrograms capture spectral shape but lose temporal texture. Hand-crafted features capture complementary information — pitch trajectory, voiced/unvoiced ratio, energy envelope — not easily derivable from 2D spectrograms alone.

Fig. 1 shows the two-branch architecture. Branch A processes a Log-Mel Spectrogram through a fine-tuned CNN; Branch B computes a 70-dimensional hand-crafted vector and projects it to 128 dimensions with batch normalisation. Both embeddings are concatenated before a Dense classifier head.

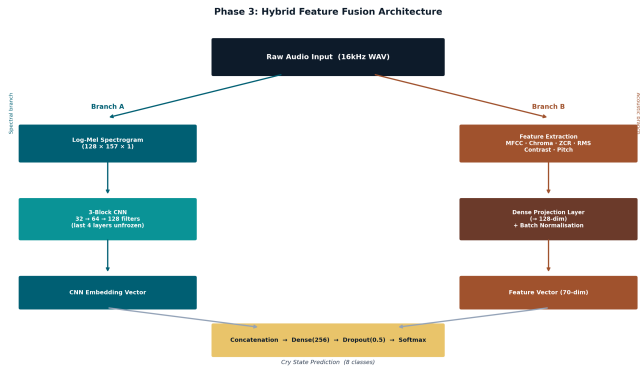


Figure 1. Phase 3 Hybrid Feature Fusion Architecture. Branch A (teal): CNN spectrogram backbone with partially unfrozen layers. Branch B (brown): 70-dim hand-crafted feature vector projected to 128 dims. Concatenated outputs pass through Dense(256)→Dropout(0.5)→Softmax.

B. Branch A: CNN Spectrogram Backbone

Branch A reuses the Phase 2 CNN with selective fine-tuning: input Log-Mel Spectrogram ($128 \times 157 \times 1$); 3-block CNN with filter progression $32 \rightarrow 64 \rightarrow 128$; last 4 layers unfrozen; output via Global Average Pooling. Freezing all but the last block preserves low-level spectral detectors learned in Phase 2 while adapting to the revised 8-class space.

C. Branch B: Hand-crafted Feature Extractor

Branch B computes a **70-dimensional** acoustic feature vector using librosa [2]. Table 2 summarises each feature group.

Table 2. 70-Dimensional Hand-crafted Feature Vector

Feature	Agg.	Dim	Role
MFCCs ($n=13$)	μ, σ	26	Timbral texture
Chroma ($n=12$)	μ, σ	24	Harmonic structure
ZCR	μ, σ	2	Voiced/unvoiced
RMS	μ, σ	2	Amplitude; Pain vs. Tired
Energy			
Sp. Contrast ($n=7$)	μ, σ	14	Tonal richness
Pitch (pip-track)	μ, σ	2	F0 trajectory; key Hungry discriminator
Total		70	

```

1 def extract_handcrafted_features(audio_path, sr
2   =16000):
3     y, _ = librosa.load(audio_path, sr=sr)
4     mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc
5       =13)
6     chroma = librosa.feature.chroma_stft(y=y, sr=sr)
7     zcr = librosa.feature.zero_crossing_rate(y)
8     rms = librosa.feature.rms(y=y)
9     contrast = librosa.feature.spectral_contrast(y=y,
10      sr=sr)
11     pitches, _ = librosa.piptrack(y=y, sr=sr)
12     pv = pitches[pitches > 0]
13     pm = pv.mean() if len(pv)>0 else 0.0
14     ps = pv.std() if len(pv)>0 else 0.0
15     return np.concatenate([
16         mfcc.mean(axis=1), mfcc.std(axis=1), # 26
17         chroma.mean(axis=1), chroma.std(axis=1), #
18         24
19         zcr.mean(axis=1), zcr.std(axis=1), # 2
20         rms.mean(axis=1), rms.std(axis=1), #
21         2
22         contrast.mean(axis=1), contrast.std(axis=1), #
23         14
24         [pm, ps], #
25         2
26     ]) # 70-dim total
    
```

Listing 1. Branch B: 70-dim Feature Extractor

D. Fusion and Classifier Head

The two branch outputs are fused via concatenation:

$$z_{\text{fused}} = [e_{\text{CNN}} \parallel W_{\text{proj}} f_{\text{craft}}] \quad (2)$$

$$\hat{y} = \text{Softmax}\left(W_2 \text{Drop}(\text{ReLU}(W_1 z_{\text{fused}} + b_1)) + b_2\right) \quad (3)$$

where e_{CNN} is the GAP output of Branch A and W_{proj} projects the 70-dim vector to 128 dimensions.

IV. Training Configuration

Table 3 summarises all training hyperparameters.

Table 3. Phase 3 Training Hyperparameters

Parameter	Value	Rationale
Optimiser	Adam	Adaptive LR
LR	5×10^{-5}	Protects frozen CNN
Loss	Sparse Cat. CE	Multi-class integer labels
Max epochs	50	Early stopping active
Early stop	patience=7	On val accuracy
Batch size	32	Standard for audio
Frozen layers	All but last 4	Preserve spectral detectors
Regularisation	L2(0.001)+Dropout(0.5)	Prevent overfitting
Class weights	balanced	Penalise minority errors
Train/test	80/20 stratified	Maintain class ratios

V. Results

A. Per-Class Classification Report

Table 4 presents the per-class precision, recall, and F1 scores on the 240-sample held-out test set.

Table 4. Phase 3 v2 Classification Report ($n=240$)

Class	P	R	F1	N
Belly Pain	0.35	0.32	0.33	25
Burping	0.29	0.32	0.30	22
Discomfort	0.12	0.19	0.14	27
Hungry	0.41	0.38	0.39	96
Laugh	1.00	1.00	1.00	22
Silence	1.00	1.00	1.00	22
Tired	0.11	0.08	0.09	26
Accuracy			0.42	240
Macro avg	0.47	0.47	0.47	240
Wtd. avg	0.44	0.42	0.43	240

B. Confusion Matrix

Fig. 2 shows the Phase 3 v2 confusion matrix. Laugh (22/22) and Silence (22/22) achieve perfect classification. The dominant error pattern is bidirectional confusion within the Hungry/Discomfort/Tired cluster: 25 Hungry samples are predicted as Discomfort, 14 Tired samples are predicted as Hungry, and 13 Discomfort samples are predicted as Hungry.

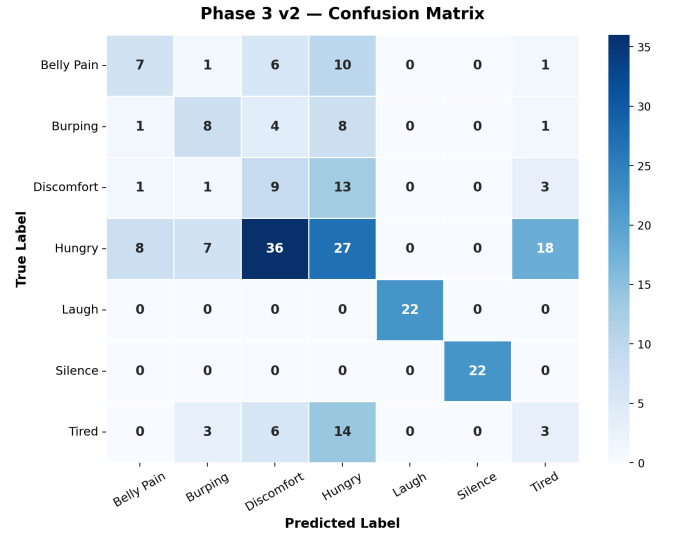


Figure 2. Phase 3 v2 Confusion Matrix (7×7). Perfect diagonal for Laugh and Silence confirms clinical-grade separability. The Hungry/Discomfort/Tired cluster shows persistent bidirectional confusion attributable to shared harmonic structure — a structural limitation of frame-averaged features.

C. Spectrogram Visual Analysis

Fig. 3 presents Log-Mel Spectrograms for one representative sample per class, drawn from the Baby Crying Sounds dataset [1].

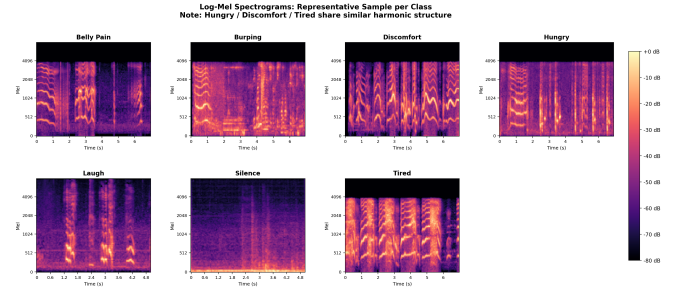


Figure 3. Log-Mel Spectrograms for representative samples of all 7 classes drawn from the Baby Crying Sounds dataset [1]. Hungry, Discomfort, and Tired share near-identical sustained harmonic stacks. Laugh exhibits distinctive high-frequency rhythmic bursts; Silence shows near-uniform low-energy distribution.

VI. Analysis

A. High-Confidence Classes

Laugh (F1 1.00) and Silence (F1 1.00) achieve perfect classification across phases. Silence has near-zero energy across all Mel bands; Laugh exhibits a distinctive high-frequency harmonic structure with unique rhythmic periodicity. Both are acoustically orthogonal to all cry states and are considered **clinically ready for deployment** in a caregiver alert system.

B. Hungry Class Recovery (+387%)

Hungry improved from F1 0.08 (Phase 3 v1) to 0.39 (Phase 3 v2), attributable to two targeted interventions:

1. **Merging cold_hot:** Of 76 Hungry test samples in Phase 3 v1, 33 were predicted as cold_hot. The merge

eliminates this confusion target entirely.

2. **Pitch trajectory features:** Fundamental frequency trajectory differs measurably between Hungry (rising melodic contour, 250–500 Hz) and Discomfort (irregular, non-tonal). The 2 pitch dimensions (mean + std) are the single highest-value addition to the feature vector.

C. Persistent Challenges

Tired (F1 0.09) and Discomfort (F1 0.14) remain the most challenging classes. All three classes in the Hungry/Tired/Discomfort cluster share low-to-moderate energy levels, overlapping rhythmic cry patterns, and overlapping fundamental frequency ranges. The key discriminating signal — *cry rhythm over time* — is permanently collapsed by frame-level mean + std aggregation, making it invisible to the current architecture. This structural limitation drives the Phase 4 transition to temporal sequence modelling.

VII. Phase Comparison

Table 5 traces the architecture evolution and Table 6 shows per-class F1 progression.

Table 5. Architecture Evolution Across Phases

	Phase 1	Phase 2	Phase 3	
Arch.	Classical ML	3-block CNN	Hybrid CNN + Fusion	
Input	Features	Spectrogram	Spec. + 70-dim	
Bottleneck	Baseline	Spectral overlap	Temporal rhythm	
Accuracy	~40%	40.42%	42.00%	

Table 6. F1 Score Progression

Class	Ph.2	v1	v2	Δ
Belly Pain	Low	0.50	0.33	−0.17
Burping	—	—	0.30	—
Discomfort	—	—	0.14	—
Hungry	0.48	0.08	0.39	+0.31
Laugh	0.98	1.00	1.00	0.00
Silence	1.00	1.00	1.00	0.00
Tired	—	—	0.09	—
Macro F1	—	0.42	0.47	+0.05

VIII. Phase 4 Roadmap

Three structural limitations motivate Phase 4:

1. **No temporal context.** Frame-averaged features collapse the temporal axis, making cry rhythm invisible — the primary Tired vs. Hungry discriminator.
2. **Confusion cluster.** Hungry/Tired/Discomfort require hierarchical decomposition beyond flat multi-class classification.

3. **Dataset scale.** 1,197 recordings from the Baby Crying Sounds dataset [1] is insufficient for robust NICU generalisation. Preterm cry recordings have shorter duration, higher F0, and weaker harmonics [5] — representing a domain shift requiring targeted data collection.

Proposed interventions: (i) LSTM over CNN frame embeddings to capture temporal cry rhythm; (ii) hierarchical 3-stage classifier: binary {Silence/Laugh} vs. {cry}, then multi-class, then specialist Hungry vs. Tired; (iii) confidence-gated routing ($p_{\max} < 0.60$ triggers specialist); (iv) data augmentation via time-stretching ($\pm 15\%$), pitch-shifting (± 2 semitones), and additive Gaussian noise to reach $\geq 3,000$ samples per class.

IX. Conclusion

Phase 3 demonstrates that Hybrid Feature Fusion augments CNN spectral representations with complementary acoustic information. Applied to the *Baby Crying Sounds* dataset [1], the 70-dimensional hand-crafted feature vector — centred on pitch trajectory and spectral contrast — resolved the spectral confusion identified in Phase 2. Overall accuracy improved to 42%, the Hungry F1 recovered from 0.08 to 0.39, and macro-average F1 reached 0.47. Laugh and Silence are clinically ready for deployment. The Tired/Discomfort cluster — a structural limitation of frame-averaged features — drives the Phase 4 transition to LSTM temporal sequence modelling.

Key Results at a Glance

Accuracy: 42% **Macro F1:** 0.47
Hungry F1: 0.08 $\rightarrow$ 0.39 (+387%)
Laugh / Silence: F1 = 1.00 (clinically ready)
Dataset: Baby Crying Sounds, Kaggle [1]

IX. References

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